

Beyond Accuracy: Explainable Trustworthy Index and Explainable Robustness Index for Evaluating CNN

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ABSTRACT

Deep neural networks often achieve high classification accuracy. Despite high accuracy, the reliability of CNN models remains a concern. Sometimes CNN models may rely on background or irrelevant features, undermining trustworthiness. Besides, with the use of perturbations, models' accuracy and object reliance may decline. Such degradation raises questions about their robustness. In this research, we have proposed two novel evaluation metrics: the **Explainable Trustworthy Index (ETI)** and the **Explainable Robustness Index (ERI)**. ETI combines accuracy with the **Localization Mask Overlap Score (LMOS)**, which measures the proportion of Grad-CAM salient pixels that fall within the ground-truth object regions. A model with high ETI clearly indicates that both its accuracy and object reliance are high and satisfactory. ERI extends this to perturbed inputs by jointly evaluating accuracy stability and LMOS stability. Like ETI, a model with high ERI indicates that both its accuracy and object focus remain stable under the use of perturbations. We have used eight CNN architectures on four datasets and nine perturbations (e.g., noise, blur, motion blur, occlusion) to calculate ETI and ERI. We have demonstrated that high accuracy sometimes does not guarantee high ETI or ERI. For instance, on the EPLI dataset, Custom CNN has achieved 99.45% accuracy but the lowest ETI (0.5922) due to poor object localization. It has also provided the lowest ERI. ETI and ERI provide a more holistic assessment of model reliability for safety-critical applications.

1. Introduction

Convolutional Neural Networks (CNNs) have achieved remarkable performance in a wide range of domains such as medical imaging, agriculture, and astronomy, where predictions support important decision-making processes [1][2][3][4]. In such applications, classification accuracy is an essential performance measure, but accuracy alone does not necessarily indicate whether a model is reliable. A CNN model is considered reliable when its predictions are strictly grounded in task-relevant visual features rather than spurious correlations. Therefore, beyond predictive accuracy, models must be quantitatively evaluated to determine whether their predictions depend on task-relevant regions rather than irrelevant background cues. Despite excellent accuracy, a model may not provide reliable predictions. Explainability methods, such as Grad-CAM [5], show that a model may rely mainly on background and irrelevant regions instead of focused objects [5, 6, 7]. This raises questions about reliability and trustworthiness.

Model behavior can shift due to real-world perturbations e.g., noise, blur, lighting shifts, compression, and occlusion. These distortions can alter both CNN accuracy and the visual regions driving predictions. Consequently, even when a model maintains relatively high accuracy under perturbations, its underlying visual explanations may still

become inconsistent [4, 5, 8]. Conventional robustness evaluation methods are primarily based on accuracy change under perturbation. They do not capture whether the model maintains consistent reliance on task-relevant regions under perturbed conditions. These observations motivate an evaluation framework that considers both predictive behavior and explanation behavior.

Explainable Artificial Intelligence (XAI) provides a framework for examining visual evidences used by CNNs for their predictions. In particular, Grad-CAM generates class-discriminative heatmaps that highlight image regions contributing to a model's prediction [5, 6, 7]. Such explanations can reveal whether a model achieves high classification accuracy by focusing substantially on background or other irrelevant regions rather than target objects. Previous studies have shown that background information and spurious correlations can influence deep neural network decisions and may lead to shortcut learning [8]. Consequently, a highly accurate model does not confirm that the predictions are made on appropriate object-level information. This creates an important distinction between predictive performance and explainable reliability. This is the motivation for evaluating a framework that combines accuracy and object focus.

We have proposed two novel evaluation metrics: Explainable Trustworthy Index (ETI) and Explainable Robustness Index (ERI) using XAI methods. Our objective is not to modify or improve CNN models, rather to provide quantitative measures for evaluating and comparing their reliability and robustness. ETI measures baseline reliability on unperturbed images by jointly evaluating classification accuracy and object-focused explanation quality. In contrast,

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ERI measures robustness to input perturbations by assessing whether accuracy and explanation quality remain consistent with clean-image behavior. In essence, ETI and ERI address two different questions: ETI asks “Whether a model is accurate and appropriately focused on the target object under normal conditions?”, whereas ERI asks “Whether these properties remain stable when the input is perturbed?”

We introduce the Localization Mask Overlap Score (LMOS) to quantify object-focused explanation quality. For each test image, we create a ground-truth mask for the target object excluding the background. We use Grad-CAM [5] to identify the image regions contributing to the model’s prediction. LMOS quantifies the proportion of the model’s decision-relevant activation that overlaps with the annotated target-object region. It is observed that in Grad-CAM visualizations, red and yellow regions are typically the most activated and therefore have the strongest influence on the model’s predictions. We integrate this principle into LMOS; a higher LMOS value indicates that a larger proportion of the model’s highlighted regions falls within the target object boundary rather than the background.

To comprehensively evaluate the proposed metrics, we conduct experiments using four public real-world datasets comprising two leaf image datasets: EPLI [9], BDML[10] and two flower image datasets: FLRS [11], FR [12]. Eight CNN architectures are trained and evaluated on the datasets. We evaluate nine perturbations, including Gaussian noise, Gaussian blur, brightness adjustment, hue shift, contrast adjustment, gamma correction, motion blur, JPEG compression, and random occlusion. For each CNN architecture, ETI is calculated on clean images, while ERI is calculated from accuracy stability and LMOS stability across the evaluated perturbations.

We empirically demonstrate the importance of combining accuracy with LMOS in ETI. For example, a custom CNN achieves 99.45% classification accuracy on the EPLI dataset, yet its mean LMOS is only 0.1900, resulting in the lowest ETI score of 0.5922 among the evaluated models. In contrast, the MobileNetV2 architecture also achieves 99.45% accuracy but obtains a substantially higher mean LMOS of 0.7765 and an ETI of 0.8855. This demonstrates that models with similar high accuracy can exhibit substantially different levels of object-focused reliance and the model with higher LMOS should be selected for real-world deployment. Importantly, a low LMOS score does not inherently denote shortcut learning or overfitting, but rather quantifies the extent to which a model’s salient activations diverge from the target object.

ERI combines *accuracy stability* and *LMOS stability* to provide a joint measure of explainable robustness. This distinction is important because predictive stability and explanation stability are not correlated. For example, on the EPLI dataset, the Custom CNN under contrast perturbation maintains relatively high accuracy stability of 0.8494 but has substantially lower LMOS stability of 0.4545. Conversely, ResNet50 under the same perturbation exhibits relatively low accuracy stability of 0.4599 but higher LMOS stability

of 0.7830. These demonstrate that accuracy stability alone cannot fully characterize robustness: a model may preserve its predictive performance while substantially changing its object-focused behavior, or it may preserve its visual focus while its predictive performance deteriorates.

The significance of ETI and ERI lies in their ability to capture key dimensions of CNN reliability and robustness that standard accuracy metrics overlook. However, a high ETI or ERI score should not be equated with overall model trustworthiness, which encompasses broader factors beyond the scope of this work. Rather, these metrics focus specifically on predictive accuracy, object-focused explanation quality, and their stability under input perturbations - offering complementary criteria for evaluating CNN models beyond conventional accuracy benchmarks.

Two questions that the proposed measures can answer - How trustworthy a model is? Trustworthy means - predictions are based on real objects in the images. How robust a model is? Robustness means - performance of the model will not degrade under perturbations.

2. Problem Definition

Let $\mathcal{D} = \{(x_i, y_i, M_i)\}_{i=1}^N$ be a dataset containing N images, where $x_i \in \mathbb{R}^{n \times m}$ is the i -th image, y_i is its ground-truth label, and M_i is the corresponding ground-truth mask of the target object. Conventional evaluation methods primarily measure the classification accuracy of a model. Let $acc \in [0, 1]$ denote the classification accuracy of a model.

Two CNNs may achieve identical accuracy while relying on substantially different image regions. In particular, a model may achieve high accuracy while its Grad-CAM activation map reveals that the activated regions concentrated outside the target object. Hence, accuracy is insufficient to distinguish models according to their object-focused explanatory behavior.

We seek to quantify the alignment between a model’s rationale and its predictions. Let H_i denote the Grad-CAM heatmap corresponding to image x_i and prediction $\hat{y}_i$. We define a scoring function $LMOS_i \in [0, 1]$ to measure the overlap ratio between the decision-relevant regions of H_i and the ground-truth object mask M_i . To indicate model trustworthiness and robustness, we address the following two sub problems in this paper.

Firstly, we want to evaluate CNN models considering both their accuracy (acc) and object-focused explanation quality (LMOS). Formally, the objective is to develop an evaluation measure T as a function of acc and LMOS: $T = \Psi(acc, LMOS)$. We address this problem by the proposed Explainable Trustworthy Index (ETI).

Secondly, we want to incorporate CNNs behavioural changes in response to perturbations in model evaluation metric. Let $\mathcal{P} = \{p_1, p_2, \dots, p_P\}$ denote a set of input perturbations. A robust CNN should not only preserve its predictive performance under any changes in $\mathcal{P}$ but also maintain a similar object-focused activation pattern similar to the

model's behavior on the corresponding clean image. Let S_{acc} and S_{LMOS} represent the overall accuracy stability (the model's ability to perform under perturbation) and LMOS stability (the model's ability to use the object-focused are for making predictions) of a model. Our second problem is to obtain an evaluation measure R that jointly captures these two forms of stability across the evaluated perturbations: $R = \Phi(S_{acc}, S_{LMOS})$. We propose Explainable Robustness Index (ERI) to address this problem.

3. System Architecture

4. Experimental Setup

4.1. Dataset

We have used four publicly available datasets in our empirical study of the proposed evaluation measures:

- **EPLI**: EPLI (introduced as EgyPLI in [9]) contains 3,588 JPG images of size 256×256 pixels. There are images of 8 species including apple, berry, fig, guava, orange, plum, persimmon, and tomato. The dataset is annotated for image-level classification [9].
- **BDML**: BDML(also known as BDMediLeaves [10]) is a dataset of 2,029 original leaf images (augmented to 38,606) from 10 medicinal plant species. They are captured with an iPhone 13 under natural light at 3024×4032 resolution. This dataset is designed for machine learning-based species identification. We used the original dataset (not the augmented one) in our experiments [10].
- **FLRS**: FLRS (or Flowers [11]) is a large image dataset for flower classification containing 3,670 training images of five flower species including daisies, tulips, sunflowers, dandelions, and roses. It is available through the TensorFlow Datasets (TFDS) collection [11].
- **KFLR**: This kaggle dataset is very similar to the FLRS dataset above consisting the same five species (daisies, dandelions, roses, sunflowers, and tulips). However, the kaggle version contains 4,242 JPG images of resolution of 320×240 pixels.

Dataset preparation. The BDML dataset is already splitted into training, validation and testing sets. We split the other three datasets into training, validation and test subsets with an approximate ratio of 80%, 10% and 10%. Table 1 presents the class-wise training, validation, and test splits. To simplify explainability and localization analysis, the test images have been manually annotated to generate corresponding JSON files. These JSON annotations describe the polygonal boundaries of the target leaf in each image. The JSON files have been converted into binary masks. Here, each mask represents only the target leaf region and excludes the background and other objects in the scene.

Table 1

Class-wise distribution of the experimental datasets.

Dataset / Class	Total	Train	Val	Test
<i>EgyPLI</i>				
Apple	519	415	52	52
Berry	340	272	34	34
Fig	508	406	50	52
Guava	520	416	52	52
Orange	547	437	55	55
Plum	468	374	47	47
Persimmon	527	421	53	53
Tomato	159	127	16	16
Subtotal	3,588	2,868	359	361
<i>Flowers (TFDS)</i>				
Daisy	633	506	64	63
Dandelion	898	718	90	90
Roses	641	512	64	65
Sunflowers	699	560	70	69
Tulips	799	639	80	80
Subtotal	3,670	2,935	368	367
<i>Flowers Recognition</i>				
Daisy	764	611	76	77
Dandelion	1052	842	105	105
Roses	784	627	78	79
Sunflowers	733	586	73	74
Tulips	984	787	98	99
Subtotal	4,317	3,453	430	434

4.2. Baseline models and evaluation metrics.

TBD

4.3. Classification Performance

4.4. Implementation of the Proposed Metrics

Both of the proposed TI and RI metrics use LMOS score. LMOS measures the consistency between model-generated explanations and ground-truth object regions. We briefly describe the implementation details of these metrics below.

4.4.1. Localization Mask Overlap Score (LMOS).

LMOS is an instance-level measure that quantifies how much the model's decision-relevant activation overlaps with the true object region. Unlike accuracy, which is a dataset-level summary, LMOS evaluates the explanation quality for each image and can be averaged across the test set. Let $H \in \mathbb{R}^{h \times w}$ denote the Grad-CAM heatmap generated from the last convolutional layer of a network. H is resized to the model's input image size. The resized heatmap size is 299×299 for Xception, 224×224 for other models. A binary attention map is obtained from H by thresholding:

$$\hat{H}(i, j) = \mathbb{I}(\text{Resize}(H)(i, j) > \tau)$$

where $\mathbb{I}(\cdot)$ is an indicator function and τ is the threshold. We set $\tau = 0.7$ to restrict LMOS calculations to the most highly activated red and yellow regions of the Grad-CAM map, which correspond to the pixels with the strongest

influence on the prediction. Hence, $\hat{H}(i, j)$ is a binary mask representing the activated decision-relevant regions.

LMOS is defined as the ratio of activated decision-relevant regions inside the object region and the total activated decision-relevant regions. Mathematically:

$$\text{LMOS} = \frac{\sum_{i=1}^h \sum_{j=1}^w \hat{h}(i, j) M(i, j)}{\sum_{i=1}^h \sum_{j=1}^w \hat{H}(i, j)} \quad (1)$$

where $M \in \{0, 1\}^{h \times w}$ denotes the resized ground-truth binary segmentation mask, and $\hat{H}(i, j)$ is the thresholded activation map.

4.4.2. Trustworthy Index (TI)

In this article, we propose Trustworthy Index (TI) to measure trustworthiness of a CNN model's predictions. We combine the accuracy, explainable consistency and the mean LMOS (Eq. 1) to define TI.

Let acc denote the classification accuracy of a model on test set and $\overline{\text{LMOS}}$ denote the mean LMOS scores across the same test samples. We define TI defined as follows:

$$\text{TI} = \frac{acc + \overline{\text{LMOS}}}{2}.$$

In this study, both acc and LMOS are equally important and are normalized to the range $[0, 1]$. acc measures predictive performance, whereas LMOS measures the mean explanation consistency with the target object. We provide equal weighting to both components to avoid favoring either predictive performance or explanatory quality.

Variations of TI. To demonstrate the impact of LMOS and acc on the proposed TI, we have calculated two variants of TI as follows:

$$\text{TI}_{\text{w/o LMOS}} = \frac{acc}{2}, \quad \text{TI}_{\text{w/o acc}} = \frac{\text{LMOS}}{2}.$$

4.4.3. Robustness Index (RI)

We estimate the robustness of a model by measure its' performance deviation under perturbation in the test datasets. Let acc_c denote the accuracy of a model on a clean test dataset (not perturbed) and acc_p denote the accuracy obtained on the test set after applying a specific perturbation p (see Sec. 4.6 for all the perturbation methods used in this research). We introduce the Accuracy Stability (S_{acc}) score under a specific perturbation p as:

$$S_{acc}^p = 1 - \frac{|acc_c - acc_p|}{acc_c + acc_p + \epsilon}$$

where ϵ is a small positive constant introduced to avoid division by zero error. A higher value of S_{acc}^p indicates that the model's performance is more consistent under the corresponding perturbed condition p .

To estimate an overall measure of S_{acc} , we compute S_{acc}^p across all $p \in P$ where P is the set of all perturbation

methods used in the experiments, see Section 4.6 and take the mean of all S_{acc}^p . This can be defined as:

$$\overline{S}_{acc} = \frac{1}{|P|} \sum_{p=1}^{|P|} S_{acc}^p$$

Similarly, we define LMOS Stability to measure how closely a model's object reliance under perturbation remains consistent with its behavior on the clean image.

For each test sample, we have computed LMOS for both the original image and its perturbed version. Let N be the number of test samples in a test set. Let $\text{LMOS}_{\text{clean}}$ denote the LMOS value computed from an original image and $\text{LMOS}_{\text{perturbed}}$ denote the LMOS value computed from the corresponding perturbed image.

To quantify the consistency between these two values, we define the LMOS stability under a perturbation:

$$S_{\text{LMOS}} = 1 - \frac{|\text{LMOS}_{\text{clean}} - \text{LMOS}_{\text{perturbed}}|}{\text{LMOS}_{\text{clean}} + \text{LMOS}_{\text{perturbed}} + \epsilon}$$

, where ϵ is a small constant to avoid division by zero error.

A higher value of S_{LMOS} indicates that the model's reliance on the focus object remains more consistent under the corresponding perturbation. The LMOS stability under a perturbation p is calculated as an average across all N test samples as:

$$S_{\text{LMOS}}^{(p)} = \frac{1}{N} \sum_{i=1}^N S_{\text{LMOS},i}^{(p)} \quad (2)$$

where $S_{\text{LMOS},i}^{(p)}$ denotes the LMOS stability of the i -th test sample under the perturbation p , and N represents the total number of test samples.

To obtain an overall measure of LMOS stability across all evaluated perturbations, the perturbation-specific LMOS stability scores are subsequently averaged as:

$$\overline{S}_{\text{LMOS}} = \frac{1}{P} \sum_{p=1}^P S_{\text{LMOS}}^{(p)} \quad (3)$$

where P denotes the total number of evaluated perturbations. In this study, $P = 9$.

To jointly evaluate the robustness of both classification performance and explainability under input perturbations, we have proposed the *Explainable Robustness Index* (ERI). Unlike accuracy stability and LMOS stability, which separately assess the consistency of predictive performance and model localization behavior, ERI provides a unified measure of robustness by combining the accuracy stability and LMOS stability.

Let s_{acc}^p be accuracy stability and s_{LMOS}^p be LMOS stability under perturbation p . Now, ERI under perturbation p is calculated as:

$$ERI^p = \frac{S_{acc}^p + S_{LMOS}^p}{2}. \quad (4)$$

Let $\bar{S}_{acc}$ denote the overall accuracy stability and $\bar{S}_{LMOS}$ denote the overall LMOS stability, both obtained by averaging their corresponding perturbation-specific stability scores across the P evaluated perturbations. The proposed ERI is defined as:

$$ERI = \frac{\bar{S}_{acc} + \bar{S}_{LMOS}}{2}. \quad (5)$$

Here, $\bar{S}_{acc}$ represents the overall consistency of classification accuracy between clean and perturbed inputs, while $\bar{S}_{LMOS}$ represents the overall consistency of the model's reliance on the focus object under perturbations. Therefore, ERI jointly captures two complementary dimensions of robustness: *predictive stability* and *explanatory stability*.

Since both component scores are bounded in the same range, the resulting ERI also provides a normalized measure of overall robustness. A higher ERI indicates that a model simultaneously maintains its classification performance and preserves its object-focused behavior across the evaluated perturbations. Conversely, a lower ERI indicates that the model exhibits greater instability in one or both dimensions. **Variations of RI.** To analyze the contribution of the individual components of RI (i.e., S_{acc} and S_{LMOS}), we calculate two variants by removing one component at a time:

$$RI_{w/o \text{ LMOS}} = \frac{\bar{S}_{acc}}{2}, \quad RI_{w/o \text{ acc}} = \frac{\bar{S}_{LMOS}}{2}.$$

4.5. Grad-CAM Visualization

Grad-CAM visualization shows that CNN models often predict correctly but they sometimes rely on the background or non-object regions. Besides, under perturbations, their reliance on object pixels hugely changes. *it is not necessary here. We may include the implementation of grad-cam in the method implementation section above*

4.6. Perturbation methods

To evaluate the robustness of the proposed framework under distribution shifts and common image corruptions, we have applied nine image perturbation methods to the test set. The perturbation parameters have been fixed across all experiments to ensure reproducibility and enable fair comparisons among different CNN architectures. The details of the perturbations are given below:

- **Gaussian Blur:** Applied using a 5×5 kernel to simulate the loss of fine spatial details caused by defocus or low-quality imaging conditions.
- **Gaussian Noise:** Additive Gaussian noise with zero mean and standard deviation 0.05 has been introduced to model sensor noise and random intensity fluctuations.

- **Brightness Adjustment:** Increased by adding a constant value of 0.2 to normalized pixel intensities, after which values have been clipped to the valid range $[0, 1]$.
- **Hue Shift:** Hue values in the HSV color space have been shifted by 15 units while preserving the saturation and value channels, simulating color variations caused by illumination or camera characteristics.
- **Contrast Adjustment:** Modified using mean-centered linear scaling: $I' = (I - \mu_I) \times 0.5 + \mu_I$, where I denotes the input image and μ_I represents the mean pixel intensity (computed per channel). The resulting values have been clipped to the range $[0, 1]$.
- **Gamma Correction:** Nonlinear illumination changes have been simulated using gamma correction: $I' = I^{2.0}$, where pixel intensities have been raised to the power of $\gamma = 2.0$.
- **Motion Blur:** Simulated using a 7×7 horizontal averaging kernel to emulate camera or object motion during image acquisition.
- **JPEG Compression:** Artifacts have been introduced by encoding and decoding the images with a quality factor of 30, simulating degradation due to lossy storage or transmission.
- **Random Occlusion:** A randomly positioned rectangular region corresponding to 20% of the image height and 20% of the image width has been masked by setting its pixel values to zero, thereby simulating partial object occlusion.

4.7. Training Configurations

We used TensorFlow/Keras on the Kaggle environment with two NVIDIA Tesla T4 GPUs for the training and evaluation of all CNN models in our experiments. The training configuration follows the settings adopted in the original EgyPLI study [9]. The custom CNN was trained from scratch, while other models were initialized with ImageNet-pretrained weights and their backbone layers were frozen.

For the pretrained models, the original classification heads were removed and replaced with a common classification head consisting of Global Average Pooling, a dropout layer with dropout rate 0.3, a fully connected layer with 128 ReLU units, and a final softmax layer configured according to the number of classes in each dataset. The Adam optimiser was used with a learning rate of 5×10^{-5} and categorical cross-entropy loss in all model optimizations.

5. Result and Discussion

5.1. Grad-CAM Visualization

5.1.1. Threshold Justification

5.2. Explainable Trustworthiness Index (ETI)

5.3. Explainable Robustness Index (ERI)

EgyPLI. XXXXXX

5.3.1. *BDMediLeaves*

5.3.2. *Flowers*

5.3.3. *Flowers Recognition KAGGLE Dataset*

5.4. Cross-Dataset Analysis

6. Related Works

7. Conclusion

Data and Code availability

Supplementary code and data (including .json files and .keras files) associated with this research are available at:

https://github.com/ragib1996/ETI_ERI

CRedit authorship contribution statement

Ragib Mehedi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – Original Draft, Writing – Review & Editing. **Azmain Inquaid Haque:** Investigation, Data Curation, Software, Validation. **Md. Nazmul Abdal:** Data Curation. **Abdullah Al Noman:** Supervision, Writing – Review & Editing. **Md. Zahidul Islam:** Supervision, Writing – Review & Editing .

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